Solution Requirements

Date	29 June 2026
Team ID	xxxx
Project Name	Opticrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Solution Requirements

The system accepts farm information such as **soil type, rainfall, temperature, humidity, pH level, and nutrient values (Nitrogen, Phosphorus, and Potassium - N, P, K). After validating the input data, the machine learning model analyzes these parameters using historical agricultural datasets and predicts the most suitable crop for cultivation. The recommended crop is then displayed through a simple and user-friendly web application, enabling farmers to make informed decisions, improve crop productivity, optimize resource utilization, reduce crop failure, and promote sustainable agricultural practices.**

.Functional Requirements

Requirement ID	Requirement	Description
FR-01	User Input	The system shall allow farmers to enter crop, soil, weather, and location details.
FR-02	Data Validation	The system shall validate the entered agricultural information before processing.
FR-03	Dataset Access	The system shall retrieve historical crop, weather, and soil data from the dataset.
FR-04	Crop Prediction	The machine learning model shall recommend the best crop and provide production optimization.
FR-05	Display Result	The system shall display the crop recommendations, expected yield, and farming tips to the user.

Non-Functional Requirements

Requirement ID	Requirement	Description
NFR-01	Performance	The system should generate crop recommendations within a few seconds.
NFR-02	Accuracy	The prediction model should provide highly accurate crop recommendations.
NFR-03	Reliability	The system should consistently provide reliable recommendations.
NFR-04	Security	User and farm data should be stored and processed securely.
NFR-05	Usability	The application should provide a simple and user-friendly interface that is easy to use.

Software Requirements

Software	Purpose
Python	Programming language
Jupyter Notebook	Model development and testing
VS Code / PyCharm	Code development
Flask	Web application framework
Pandas	Data manipulation
NumPy	Numerical computations
Scikit-learn	Machine learning algorithms
Matplotlib	Data visualization
Seaborn	Statistical visualization

Hardware Requirements

Component	Specification
Processor	Intel i3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	Minimum 2 GB Free Space
Operating System	Windows / Linux / macOS
Internet	Required for dataset download and deployment